

Minimization of Maximum Mean Discrepancy (MMD)

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Annual student conference, UCL Computer Science, 2026

About Me



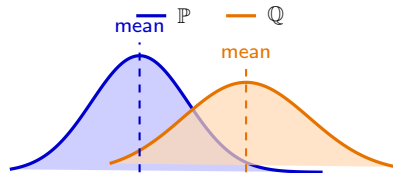
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- 4th year PhD Student at University College London (UCL)
 - Foundational AI Centre
 - Gatsby Computational Neuroscience Unit
- Graduated from Tsinghua University in 2022
 - Department of EE
- Kernel (nonparametric) methods, causal inference, statistical learning theory

Content: Minimization of MMD

- What is MMD?
- Minimization of MMD for numerical integration
- Minimization of MMD for sampling / generative modelling
- Minimization of MMD for statistical inference

Background: what is MMD?

- Given two distributions $\mathbb{P}$ and $\mathbb{Q}$, how to compare them?
- Compute the difference of their means: $|\mathbb{E}_{\mathbb{P}}[X] - \mathbb{E}_{\mathbb{Q}}[X]|$.
- Compute the difference of their second moments: $|\mathbb{E}_{\mathbb{P}}[X^2] - \mathbb{E}_{\mathbb{Q}}[X^2]|$.
- The first two moments are not enough!



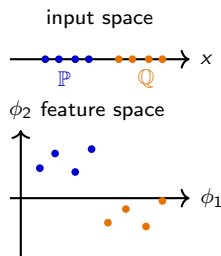
Goal: Compare $\mathbb{P}$ and $\mathbb{Q}$ with a general f , i.e., $|\mathbb{E}_{\mathbb{P}}[f(X)] - \mathbb{E}_{\mathbb{Q}}[f(X)]|$?

Background: Maximum Mean Discrepancy (MMD)

- Compare $\mathbb{P}$ and $\mathbb{Q}$ by maximizing the difference over a function class $\mathcal{H}$:

$$\text{MMD}(\mathbb{P}, \mathbb{Q}) := \sup_{f \in \mathcal{H}} |\mathbb{E}_{\mathbb{P}}[f(X)] - \mathbb{E}_{\mathbb{Q}}[f(X)]|.$$

- $\mathcal{H}$ is a reproducing kernel Hilbert space (RKHS) with kernel k .
- The RKHS contains functions that are linear combinations of the feature maps $\phi : \mathcal{X} \rightarrow \mathcal{H}$, $f = \sum_{i=1}^n \alpha_i \phi(x_i)$.



Theorem 1

MMD admits a closed-form expression:

$$\text{MMD}^2(\mathbb{P}, \mathbb{Q}) = \mathbb{E}_{X, X' \sim \mathbb{P}}[k(X, X')] + \mathbb{E}_{Y, Y' \sim \mathbb{Q}}[k(Y, Y')] - 2\mathbb{E}_{X \sim \mathbb{P}, Y \sim \mathbb{Q}}[k(X, Y)].$$

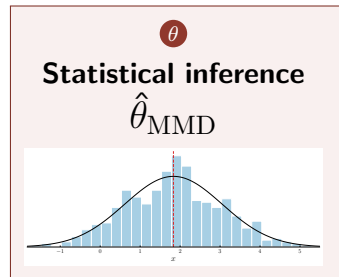
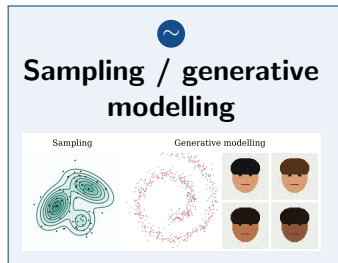
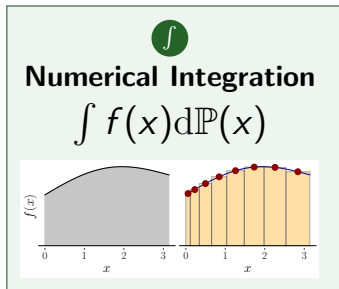
- MMD is a probability divergence ($\text{MMD}(\mathbb{P}, \mathbb{Q}) = 0 \iff \mathbb{P} = \mathbb{Q}$) if k is characteristic (Gaussian).

Background: Maximum Mean Discrepancy (MMD)

We now have MMD

A kernel-based way to compare probability distributions $\mathbb{P}$ and $\mathbb{Q}$

What can we use it for?

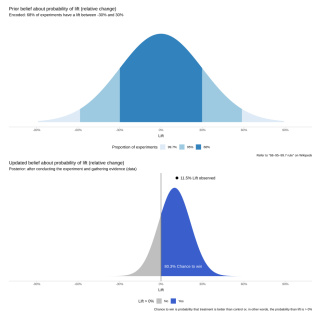


Part I: Numerical Integration

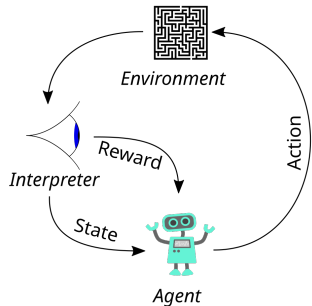
Where do integrals appear in computer science?



Computer graphics
Rendering



Bayesian inference
Posterior expectations



Reinforcement learning
Expected returns

Part I: Numerical Integration

- Given a function f and a distribution $\mathbb{P}$, how to compute $\int f(x)d\mathbb{P}(x)$?
- Given a set of points $\{x_i\}_{i=1}^n$ and function evaluations $\{f(x_i)\}_{i=1}^n$.
- Quadrature rule:

$$\hat{I} = \sum_{i=1}^n w_i f(x_i).$$

- When $w_1 = \dots = w_n = 1/n$, this is Monte Carlo. $\hat{I}_{MC} = \frac{1}{n} \sum_{i=1}^n f(x_i)$.

$$\mathbb{E} \left[|\hat{I}_{MC} - I| \right] = \mathcal{O}(n^{-1/2}).$$

- Or equivalently, to obtain an ϵ -accurate estimate, we need $n = \mathcal{O}(\epsilon^{-2})$ samples.

Question: Can we do better than $\mathcal{O}(n^{-1/2})$?

Answer: Find better weights $w_1, \dots, w_n$.

Part I: Numerical Integration

- Suppose $f \in \mathcal{H}$ is in the RKHS associated with k .
- Denote $\hat{\mathbb{P}} = \sum_{i=1}^n w_i \delta_{x_i}$ as the weighted empirical distribution.

$$\left| I - \hat{I} \right|^2 = \left| \int f(x) d\mathbb{P}(x) - \sum_{i=1}^n w_i f(x_i) \right|^2 \leq \|f\|_{\mathcal{H}}^2 \cdot \text{MMD}^2(\hat{\mathbb{P}}, \mathbb{P}).$$

- The right-hand side is the **worst-case** error of the quadrature rule.
- The **optimal** weights that minimize $\text{MMD}(\hat{\mathbb{P}}, \mathbb{P})$:

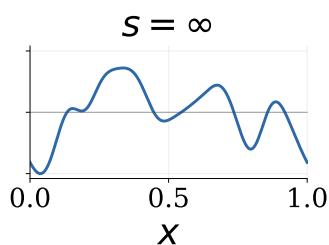
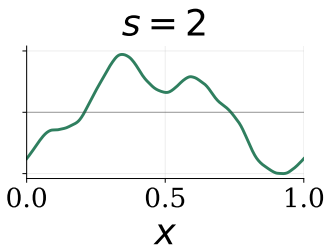
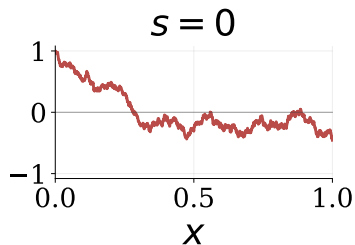
$$w_{1:n} = \mathbb{E}_{X \sim \mathbb{P}}[k(X, x_{1:n})] K(x_{1:n}, x_{1:n})^{-1}.$$

- The kernel quadrature (KQ) rule: $\hat{I}_{\text{KQ}} = \sum_{i=1}^n w_i f(x_i)$.

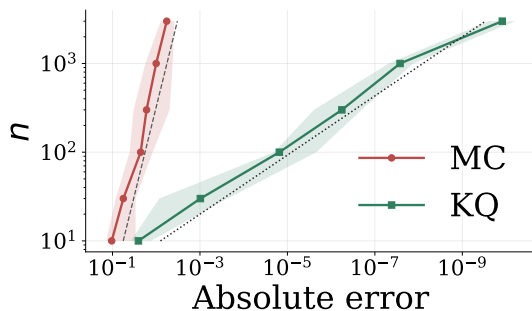
Part I: Numerical Integration

Rate: KQ achieves $\mathcal{O}(n^{-s/d})$ rate.

- s is the smoothness of f , d is the dimension of input x .
- Better than $\mathcal{O}(n^{-1/2})$ when $s > d/2$.



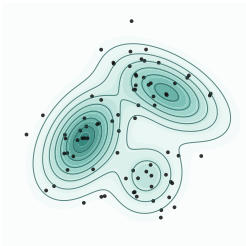
Part I: Numerical Integration



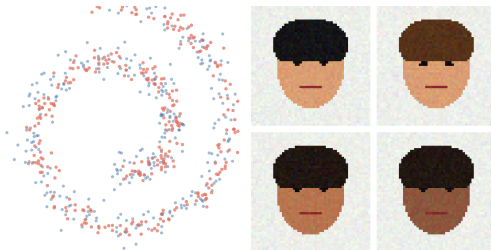
- 1 **Nested Expectations with Kernel Quadrature.** [Z. Chen](#), M. Naslidnyk, F.-X. Briol ([ICML 2025](#))
- 2 **Conditional Bayesian Quadrature.** [Z. Chen](#), M. Naslidnyk, A. Gretton, F.-X. Briol ([UAI 2025](#))

Part II: Sampling / Generative modelling

Sampling



Generative modelling



- Given a target distribution $\mathbb{Q}$, how to draw samples $\{x_i\}_{i=1}^n$ from it?
 - When $\mathbb{Q}$ is known up to normalization, this is sampling.
 - When $\mathbb{Q}$ is known up to i.i.d samples, this is generative modelling.

Problem: How to learn a target probability distribution $\mathbb{Q}$ on $\mathbb{R}^d$.

- This problem can be written as an **optimization** problem on $\mathcal{P}(\mathbb{R}^d)$, the space of probability distributions.

$$\arg \min_{\mathbb{P} \in \mathcal{P}(\mathbb{R}^d)} D(\mathbb{P} \mid \mathbb{Q}).$$

- Here D is a similarity metric or distance, e.g. Kullback–Leibler divergence / MMD.
 - $D(\mathbb{P}_1 \mid \mathbb{P}_2) = 0$ if and only if $\mathbb{P}_1 = \mathbb{P}_2$.
- How to find the minimum? Gradient descent!

Part II: Sampling / Generative modelling

Question: Wait... How to define gradients on $\mathcal{P}(\mathbb{R}^d)$?

Answer: Wasserstein gradient !

../plotting/figures/ot_wasserstein_gradient_illustration.pdf.pdf